Same Recipe, Different Fate: The Training Recipe's Effect on Safety Erosion from Benign Fine-Tuning Depends on the Model Family

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Abstract

Benign fine-tuning can erode safety, but reported effects vary widely. We compare six instruct models from OLMo-2, Llama-3 and Qwen2.5, at about 1B and at 7–8B, using LoRA on 4,800 Dolly rows with three seeds per main recipe cell. Erosion E is the rise in Qwen3Guard's Unsafe share of 600 sampled answers to 200 development harm prompts; HarmBench is a second judge. At 1B, a Scheduled recipe (warmup, cosine decay and token-mean loss) lowers OLMo-2-1B's mean erosion from $+45.1$ to $+28.8$ pp, compared with $+59.3$ to $+57.1$ for Llama-3.2-1B and $+71.9$ to $+73.9$ for Qwen2.5-1.5B. Both OLMo family-interaction intervals exclude zero on both judges, but OLMo's over-refusal rises by 11.7 pp. The pre-registered OLMo-versus-Llama replication at 7–8B is inconclusive. Qwen2.5-7B's mean erosion falls from $+78.1$ to $+68.9$ pp; its interaction with OLMo excludes zero on Qwen3Guard only. Halving the learning rate lowers mean erosion in OLMo-7B and Llama-8B, without establishing equal effects. Under one recipe, OLMo erodes least and Qwen most at both sizes. Dark–light fiction differences lie within wide pre-registered margins. A blind human read suggests the Unsafe label mostly tracks lost refusal. All results are development evidence; we report every screened setting and the judges' limits.

1 Introduction

Fine-tuning an aligned language model on benign data can erode its safety (Qi et al., 2024; He et al., 2024). After learning from ordinary examples, the model refuses fewer harmful requests. Small open models are often fine-tuned on users' own data, so we need to understand how much safety they lose, and which training choices change that loss.

Published numbers differ widely, and each study uses its own model family, recipe, data and judge. Family contrasts exist (Hawkins et al., 2026; Khan et al., 2026; Guo et al., 2026), and optimisation choices drive much of the safety loss (Kim et al., 2025). To understand the differences, we need to compare families and recipes together, using the same pipeline and judge.

We wanted to understand which parts of a recipe change erosion, and whether the pattern holds for different sizes of the same family. Does a recipe that helps one model also help another? We compare recipes within each model and models under matched recipes, including cases where transfer remains unestablished.

We fine-tune six instruct models from three families (OLMo-2, Llama-3 and Qwen2.5, at about 1B and 7–8B parameters) on the same 4,800 Dolly rows with one LoRA recipe, three seeds each. Then we change the loss weighting and learning-rate schedule, testing transfer across families and sizes with pre-registered new runs. A separate pre-registered test replaces Dolly with dark or light fiction of the same format and length. The erosion E is the rise in the share of sampled answers to harm prompts that a safety judge labels Unsafe.[1]

We make four contributions:

- **The recipe's effect depends on the family.** At 1B, a standard learning-rate schedule and loss cut OLMo's erosion by about a third, but raise over-refusal by 11.7 pp over Constant. Llama's and Qwen's mean erosion changes little. Both interactions exclude zero on two judges (§4, §5), and a loss-weighting-only recipe points to the schedule block as the larger contributor.

- **The 7–8B replication is inconclusive.** In a pre-registered replication at 7–8B, the OLMo-versus-Llama interaction is not shown, and its point estimate has the opposite sign on

[1] All numbers come from development prompt sets; a held-out set is kept for the larger study (§2).

both judges. Qwen's mean erosion falls most, with an interaction against OLMo on the main judge only. Half LR lowers OLMo-7B's and Llama-8B's means; equal effects are not established (§5).

- **Family ordering under a matched recipe.** Under Constant, OLMo-2 has the lowest mean erosion and Qwen2.5 the highest at both tested sizes (§3). We are not aware of earlier benign-fine-tuning erosion numbers for OLMo-2 Instruct (§10).

- **Content of benign data.** In a pre-registered test on OLMo-2-1B, dark–light differences lie within wide pre-registered margins. Both eroded far less than Dolly at the same format and length, while refusing more benign prompts (§8).

We also look at where erosion appears. About half of OLMo's rise comes from answers that no longer open with a refusal (§6), and exploratory probes find greater retention of OLMo-2-1B's harmful–harmless separation proxy under Scheduled (§7). Section 9 lists every screened setting and the judges' limits.

Our claims hold only for these models, this data, this prompt set and these judges. A rise in the Unsafe share alone does not show harmful compliance. We do not claim a cause for the family difference.

2 Setup

Models. We use six instruct models from three families, at two sizes: OLMo-2-0425-1B-Instruct and OLMo-2-1124-7B-Instruct (Team OLMo et al., 2024), Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct (Qwen et al., 2024), and Llama-3.2-1B-Instruct and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). We write OLMo-2-1B, OLMo-2-7B, Llama-3.2-1B and Llama-3.1-8B. Section B gives the revision hashes. We call each model before our fine-tune the *untuned* model.

Data. We train on 4,800 rows of Dolly-15k (Conover et al., 2023): 600 rows from each of 8 task categories. Every run is one epoch, except the High-dose screen below. We tokenise the rows separately for each family, so the same rows give different token counts.

Recipes. Every recipe trains LoRA adapters (Hu et al., 2022) of rank 64 and alpha 128 on all linear layers, with AdamW. Training runs in our own PyTorch loop, with Hugging Face Transformers and PEFT, and Table 6 lists every setting. The *Constant* recipe uses a constant learning rate of 10^{-4}, no warmup, 128 rows per step and one epoch (38 steps). Its loss is a per-row mean: each row weighs the same, so the tokens of short answers weigh more. The *Scheduled* recipe is the Constant recipe plus a 4-step linear warmup with cosine decay from a peak learning rate of 10^{-4}, and a token-mean loss, so each response token weighs the same. The Constant recipe is not standard practice; the Scheduled recipe is the standard-practice comparison. Each variant changes one thing: *Constant, token-mean* changes only the loss weighting; *Constant, half LR* and *Scheduled, half LR* halve the (peak) learning rate to 5×10^{-5}; *Constant, rank 16* and *Scheduled, rank 16* use rank 16 and alpha 32. The 7–8B models use the same recipes with 2 rows per micro-batch and 64 accumulation steps, so the same 128 rows per step. On Qwen2.5-1.5B only, we first screened a *High-dose* recipe: a constant learning rate of 2×10^{-4}, 8 rows per step (600 steps per epoch) and 3 epochs.

Evaluation. The 200 development harm prompts come from HarmBench's standard behaviours (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024), SimpleSafetyTests (Vidgen et al., 2024) and the toxic set of OR-Bench (Cui et al., 2025). For each prompt, we sample 3 answers at temperature 0.7 (top-p 1.0), and we take one greedy answer. Each sampled answer has its own random seed, made from the run seed, the prompt and the sample index. An answer may have up to 2,048 new tokens. Over-refusal uses 100 benign prompts: 50 safe prompts from XSTest (Röttger et al., 2024) and 50 from OR-Bench-Hard (Cui et al., 2025). It is the share of the validly judged greedy answers to these prompts that Qwen3Guard (below) labels as a refusal (*Refusal: Yes*). Task skill is the held-out loss on 200 Dolly rows. We also report ARC forced choice on 189 questions (Clark et al., 2018).

Judges and erosion. The main judge is Qwen3Guard-Gen-4B (Zhao et al., 2025a). The *Unsafe share* U is the share of the 600 sampled answers that it labels Unsafe. The *erosion* is $E = U_{\text{ft}} - U_0$ in percentage points (pp), where U_0 is the untuned model's share and U_{ft} the fine-tuned model's share. U is a broad policy label, not an attack success rate (§9). Granite Guardian 3.3 8B (Padhi et al., 2024), in its 4-bit Q4_K_M build

with its harm criterion, is a second judge on the 200 greedy answers. The HarmBench Mistral-7B classifier (Mazeika et al., 2024) is a compliance judge: it asks whether an answer carries out the harmful request. As in HarmBench's code, it reads at most the first 511 tokens of an answer. We write its erosion as E_{HB}. Section B also gives the judges' revisions.

Statistics. We cluster the standard error (SE) of U by prompt, because the 3 answers to one prompt are correlated; the design effect is about 2. The band for E is $E \pm 2\,\mathrm{SE}$. It covers evaluation noise only. With 3 seeds, the seed standard deviation (SD) has 2 degrees of freedom (df).

Development prompts only. Every number in this note comes from the development prompt sets above; the larger study keeps a separate held-out set, which we did not use. So no result here is confirmed on prompts outside the development set, and some choices, such as the partial window (§3), were made after seeing development results (§9).

Release. We will release the recipe files, the per-answer judge labels (no prompt or answer text), the analysis scripts and the figure data with the final version.

3 Same recipe, different fate

Table 1 gives the erosion under the Constant recipe, three seeds per model. OLMo-2-1B rises from 3.2% to a mean of 48.3%: the mean E is $+45.1$ pp (band 41.3–49.0). Llama-3.2-1B rises by $+59.3$ pp and Qwen2.5-1.5B by $+71.9$ pp. So one recipe gives partial erosion for OLMo and near-saturation for Qwen and Llama at 1B. The 7–8B models keep this order: OLMo-2-7B rises by $+44.1$ pp, Llama-3.1-8B by $+60.7$ pp and Qwen2.5-7B by $+78.1$ pp. The seeds spread more at 7–8B (sample SDs of 6–8 pp, against 3–5 pp at 1B). We call erosion *partial* if E lies between 20 and 55 pp, the range that the larger study needs (§1). We fixed this window after the first Qwen results (§9). Both OLMo means lie inside it; the Llama and Qwen means lie above it. Figure 2 shows every seed of the 1B models.

The starting points differ: Llama-3.2-1B starts at 13.8%. So we also report the share of the headroom that each model uses, $E/(100 - U_0)$ (Table 1); the order of the models does not change. Over-refusal falls in every family. Qwen2.5-1.5B starts at 68% and Qwen2.5-7B at 7%, and both near saturation, so a high starting over-refusal does not by itself explain Qwen's near-saturation. Held-out loss falls in every run, so each model learns the task. The compliance judge keeps the order of the families: E_{HB} is $+30.3$ pp for OLMo-2-1B, $+37.9$ for Llama-3.2-1B and $+45.4$ for Qwen2.5-1.5B at 1B, and $+33.1$, $+44.4$ and $+60.6$ pp at 7–8B (Table 1). It sees 0.63–0.78 times the Qwen3Guard erosion in every model. This ratio of marginal changes does not measure the fraction of lost refusals that became harmful compliance (§9).

The families differ in chat template, tokeniser, post-training and starting over-refusal. We do not claim a cause for the family difference.

4 The recipe matters

The Scheduled recipe changes the loss weighting and the learning-rate schedule, with the other settings kept the same. On OLMo, mean erosion falls from $+45.1$ to $+28.8$ pp, with 3 seeds of each recipe (Table 2). On seed 1101, whose two runs share their evaluation sampling streams, the difference is -14.5 pp, about 5 times the between-seed SD under Constant (2.65 pp, 2 df). But over-refusal also rises (a mean of 16.3%, against 17% untuned and 4.7% under Constant), and held-out loss is a little higher (1.613 against 1.545). Its three-seed band is 25.3–32.4 pp ($E \pm 2$ SE, evaluation noise only), and Granite agrees in direction on every seed (mean greedy E $+22.3$ against $+43.0$ pp). On the HarmBench compliance judge, with every OLMo unit judged on one GPU type, the mean E_{HB} is $+30.3$ pp under the Constant recipe and $+17.2$ pp under the Scheduled recipe: the Scheduled recipe keeps 0.57 of the Constant recipe's erosion on this judge, against 0.64 on Qwen3Guard. Under Scheduled, rank 16, the erosion is $+20.2$ pp on the same seed (8.8 pp lower); by a rule fixed before the run, rank matters little, but one seed cannot exclude a real difference of this size.

Which recipe change accounts for the reduction? Constant, token-mean changes only the loss weighting. Over three seeds it gives $E = +41.9$ pp ($+43.7$, $+39.5$ and $+42.5$), against $+45.1$ pp under the Constant recipe and $+28.8$ pp under the Scheduled recipe. So, on the main measure, the loss weighting accounts for about a fifth of the cut (3.2 of 16.3 pp; the Welch SE of that difference is 2.0 pp, so its share is not distinguishable from zero), and the schedule for the rest: a 4-step warmup and a cosine decay, which also halves the mean

Table 1: Same recipe, different fate. Every row uses the Constant recipe for one epoch, three seeds per model; Section A gives every seed. Values are means over the seeds. U_0 and U_{ft} are the Unsafe shares before and after fine-tuning. $\pm$ is the sample SD of E across seeds.

Model	Seeds	U_0	U_{ft}	E (pp)[a]	E_{HB} (pp)[b]	Over-refusal, % (0 → ft)	Held-out loss (0 → ft)
OLMo-2-1B	3	3.2%	48.3%	$+\mathbf{45.1} \pm 2.6$	+30.3	17 → 4.7	2.100 → 1.545
OLMo-2-7B	3[e]	1.8%	45.9%	$+\mathbf{44.1} \pm 6.4$	+33.1	27 → 2.3	2.121 → 1.355
Qwen2.5-1.5B	3[e]	4.7%	76.6%	$+\mathbf{71.9} \pm 3.0$	+45.4	68[c] → 2.3	1.890 → 1.462
Qwen2.5-7B	3[e]	4.8%	82.9%	$+\mathbf{78.1} \pm 4.4$	+60.6	7 → 2.7	3.335 → 1.295[d]
Llama-3.2-1B	3	13.8%	73.1%	$+\mathbf{59.3} \pm 4.6$	+37.9	11 → 3.0	2.243 → 1.554
Llama-3.1-8B	3[e]	5.3%	66.1%	$+\mathbf{60.7} \pm 8.2$	+44.4	17 → 1.0	2.109 → 1.323

[a]Headroom used, $E/(100 - U_0)$: OLMo-2-1B 0.47, OLMo-2-7B 0.45, Qwen2.5-1.5B 0.75, Qwen2.5-7B 0.82, Llama-3.2-1B 0.69, Llama-3.1-8B 0.64. [b]HarmBench on A100 pods, each fine-tuned unit against its own model's untuned unit (untuned shares: OLMo-2-1B 3.2%, OLMo-2-7B 1.2%, Qwen2.5-1.5B 3.2%, Qwen2.5-7B 5.3%, Llama-3.2-1B 11.0%, Llama-3.1-8B 5.5%). [c]The 68% untuned over-refusal is Qwen2.5-1.5B only; Qwen2.5-7B starts at 7% and still nears saturation. [d]The untuned 7B held-out loss is above the 1.5B's on the same rows, so we do not compare task loss across sizes. [e]Run on a later engine identity than the OLMo-2-1B and Llama-3.2-1B rows (Section B); seed 1101 of each Qwen model reproduced the earlier one-seed unit's counts exactly.

learning rate over the epoch. HarmBench gives the same split: E_{HB} is +27.7 pp under Constant, token-mean (three seeds), so the loss weighting accounts for 2.6 of the 13.2 pp cut on this judge too. We had predicted the opposite before the first token-mean run ($E \leq 36.25$ pp, pre-registered); the three-seed mean also fails that prediction. On the 200 greedy answers the loss weighting matters more (greedy E +33.3 against +41.5 pp; Granite +34.7 against +43.0 pp; three seeds each), and the opening lexicon agrees with the main measure (§6): the schedule keeps refusal openings, and the loss weighting mainly changes answer length. This agrees with two earlier findings: optimisation choices drive much of the safety loss (Kim et al., 2025), and a warmup alone cut erosion on benign data (Qi et al., 2025).

On Llama-3.2-1B, the Scheduled recipe barely moves the erosion: a mean of +57.1 pp over three seeds (SE 2.55), against +59.3 pp under the Constant recipe. Granite agrees in direction on every seed. So the Scheduled recipe cuts OLMo's erosion more than Llama's: the change is −16.3 pp for OLMo (−0.69 on the logit scale) and −2.2 pp for Llama (−0.11). By the fixed test below, the recipe effect differs between OLMo and Llama. So the family contrast survives the standard recipe: OLMo stays partial and Llama stays near saturation. The compliance judge shows the same pattern: the Scheduled recipe keeps 0.57 of OLMo's E_{HB} (+30.3 → +17.2 pp) but 0.94 of Llama's (+37.9 → +35.8 pp; three seeds each). Scheduled, half LR (a peak learning rate of 5×10^{-5}) moves OLMo from +28.8 to +20.8 pp (two seeds) and Llama from +57.1 to +41.2 pp (one seed): about −28% in both families (−0.40 and −0.69 on the logit scale). So the learning rate moves both families, while the Scheduled recipe moves OLMo far more than Llama. HarmBench did not judge the 1B half-LR runs; §5 repeats the comparison at 7–8B, pre-registered.

The logit scale and our claim rule. Near saturation, a fixed effect on the logit scale gives a smaller change in percentage points, so we compare recipes on both scales. The test takes the logit of U for each seed. For each family, it takes the mean under the Scheduled recipe minus the mean under the Constant recipe, with a variance from the seeds. The interaction is OLMo's change minus Llama's, with a Welch interval. We say that the recipe effect differs between the families only if the 95% interval excludes 0; otherwise we describe a pattern. The interaction is −0.57 (SE 0.17, Welch df 5.1; 95% interval −1.02 to −0.13). It excludes 0, so the Scheduled recipe cuts OLMo's erosion more than Llama's. In percentage points, the contrast is −14.1 pp (SE 3.7), and its interval also excludes 0. Limits go with this result. We wrote the exact test after 11 of the 12 values were known; the design (3 seeds of each recipe in both families, on the logit scale) was in our draft before Llama's Constant seeds 1102 and 1103 ran. The interval rests on about 5 df. Its judge is Qwen3Guard. The same test on the HarmBench labels, run once HarmBench had judged all twelve units, gives −0.60 (SE 0.14, Welch df 5.6; 95% interval −0.95 to −0.25), or −11.0 pp (−18.3 to −3.7), so the second judge agrees; Granite agrees in direction on every seed.

Table 2: The recipe matters. One line per model and recipe, with means over the seeds; Section A gives every seed. $\pm$ is the sample SD of E across seeds (3-seed rows only). Over-refusal is in %. Untuned values are in Table 1; §2 defines the recipes. The rows below the second rule are the pre-registered cells of §5.

Model	Recipe	Seeds	E (pp)	E_{HB} (pp)[a]	Over-refusal	Held-out loss	Granite E (pp)[b]
OLMo-2-1B	Constant	3	$+45.1 \pm 2.6$	$+30.3$	4.7	1.545	$+43.0$
OLMo-2-1B	Constant, token-mean	3	$+41.9 \pm 2.2$	$+27.7$	7.3	1.589	$+34.7^{c}$
OLMo-2-1B	Scheduled	3	$+\mathbf{28.8} \pm 2.3$	$+17.2$	**16.3**	1.613	$+22.3$
OLMo-2-1B	Scheduled, half LR	2	$+20.8$	—	20.0	1.635	$+11.8$
OLMo-2-1B	Scheduled, rank 16	1	$+20.2$	—	25	1.649	$+9.0$
Llama-3.2-1B	Constant	3	$+59.3 \pm 4.6$	$+37.9$	3.0	1.554	$+62.8^{c}$
Llama-3.2-1B	Scheduled	3	$+57.1 \pm 2.6$	$+35.8$	7.3	1.615	$+56.7$
Llama-3.2-1B	Scheduled, half LR	1	$+41.2$	—	4	1.642	$+38.0$
Qwen2.5-1.5B	Constant	3	$+71.9 \pm 3.0$	$+45.4$	2.3	1.462	$+80.2^{c}$
Qwen2.5-1.5B	Scheduled	3	$+73.9 \pm 0.7$	$+52.7$	6.7	1.512	$+81.8^{c}$
OLMo-2-7B	Constant	3	$+44.1 \pm 6.4$	$+33.1$	2.3	1.355	$+44.8^{c}$
OLMo-2-7B	Scheduled	3	$+50.3 \pm 8.1$	$+37.8$	6.3	1.411	$+47.2^{c}$
OLMo-2-7B	Constant, half LR	3	$+\mathbf{31.9} \pm 1.4$	$+24.0$	5.3	1.370	$+35.0^{c}$
Llama-3.1-8B	Constant	3	$+60.7 \pm 8.2$	$+44.4$	1.0	1.323	$+59.3^{c}$
Llama-3.1-8B	Scheduled	3	$+59.2 \pm 3.3$	$+42.5$	5.3	1.364	$+58.5^{c}$
Llama-3.1-8B	Constant, half LR	2	$+53.3$	$+37.0$	2.0	1.327	$+52.5^{c}$
Qwen2.5-7B	Constant	3	$+78.1 \pm 4.4$	$+60.6$	2.7	1.295	$+84.7^{c}$
Qwen2.5-7B	Scheduled	3	$+\mathbf{68.9} \pm 1.9$	$+55.6$	5.7	1.336	$+73.7^{c}$

—: HarmBench did not judge these runs. [a]HarmBench on A100 pods, every seed, each unit against its model's untuned unit (shares in Table 1); units judged in two pod runs got the same labels. [b]Granite on the 200 greedy answers, against the untuned unit judged on the same device; [c]on the pod (Mac-versus-pod parity passed; untuned pod shares in Section A). On the Mac, Llama-3.2-1B's seed 1101 gives $+67.5$.

§5 applies the same test, pre-registered, to a third family and to the 7–8B models.

5 Does the recipe effect transfer across sizes and families?

Does the OLMo-versus-Llama interaction hold at a larger size, and does a third family, Qwen, behave like OLMo or like Llama? We answered both with pre-registered new runs. Their comparator cells were partly or fully known; before the new runs, we fixed the cells, test, labels and predictions in four files, with the test script hashed and self-tested on the 1B counts (Section B): (i) a 7–8B replication of the OLMo-versus-Llama test, 3 seeds per cell; (ii) the Scheduled recipe on Qwen2.5-1.5B, with T1 (OLMo-1B against Qwen-1.5B) and T2 (Qwen-1.5B against Llama-1B); (iii) the same on Qwen2.5-7B, with T5 (OLMo-7B against Qwen-7B) and T6 (Qwen-7B against Llama-8B); and (iv) the Constant recipe at half its learning rate on OLMo-2-7B (3 seeds) against Llama-3.1-8B (2 seeds). Every test is the test of §4 (Table 3), also run on the HarmBench labels. Six tests ran on overlapping cells with no multiplicity adjustment, and every unit is reported. Table 2 gives the cells.

At 1B, OLMo has the largest mean reduction. Qwen2.5-1.5B goes from $+71.9$ to $+73.9$ pp under the Scheduled recipe ($D = +0.11$). T1 reads "differs: OLMo's change is more negative" on both judges. T2 is "not shown" on Qwen3Guard but "differs" on HarmBench, where Qwen's E_{HB} rises from $+45.4$ to $+52.7$ pp while Llama's falls by 2 pp: so Qwen-1.5B's small rise is judge-dependent. We had predicted that Qwen behaves like Llama (D between -0.4 and 0; mean E between $+55$ and $+75$ pp), T1 "differs" and T2 "not shown": the mean is inside the range, D is just above it, and both test predictions held.

The 7–8B replication criterion is not met. Under the Scheduled recipe, OLMo-2-7B's mean E rises from $+44.1$ to $+50.3$ pp ($D = +0.25$) and Llama-3.1-8B's falls from $+60.7$ to $+59.2$ (-0.08). The interaction is "not shown" on both judges, with the opposite sign to 1B. We had predicted that it replicates and that OLMo-2-7B's Scheduled mean is about $+29$ pp (range $+15$ to $+40$): both failed; the Llama prediction (about $+65$, range $+55$ to $+75$) held. Granite agrees in direction with Qwen3Guard on every cell mean. The seeds spread more at 7–8B, so this test has less power than the 1B test. Its interval allows either sign: the point estimate reverses, but this does not establish absence of an interaction or a size effect.

Table 3: The pre-registered tests, and the 1B test of §4. D is a family's mean logit U under the changed recipe minus under the Constant recipe (Qwen3Guard); $I = D_{\text{first}} - D_{\text{second}}$ with its Welch 95% interval, on Qwen3Guard and on the HarmBench labels (whose own D differ). "Differs" names the family whose change is more negative when the interval excludes 0; otherwise "not shown".

Test	First − second	D_{first}	D_{second}	I [95%], Qwen3Guard	I [95%], HarmBench	Label
1B test (§4; written late)	OLMo-1B − Llama-1B	−0.69	−0.11	−0.57 [−1.02, −0.13]	−0.60 [−0.95, −0.25]	differs (OLMo)
T1	OLMo-1B − Qwen-1.5B	−0.69	+0.11	−0.80 [−1.13, −0.46]	−0.98 [−1.30, −0.65]	differs (OLMo)
T2	Qwen-1.5B − Llama-1B	+0.11	−0.11	+0.22 [−0.24, +0.69]	+0.38 [+0.08, +0.67]	not shown; HarmBench differs
Replication, 7–8B	OLMo-7B − Llama-8B	+0.25	−0.08	+0.33 [−0.49, +1.16]	+0.28 [−0.39, +0.96]	not shown
T5	OLMo-7B − Qwen-7B	+0.25	−0.57	+0.82 [+0.08, +1.56]	+0.42 [−0.09, +0.93]	differs (Qwen); HarmBench not shown
T6	Qwen-7B − Llama-8B	−0.57	−0.08	−0.48 [−1.27, +0.30]	−0.14 [−0.80, +0.53]	not shown
Half LR, 7–8B	OLMo-7B − Llama-8B	−0.51	−0.34	−0.17 [−0.95, +0.60]	−0.13 [−0.79, +0.53]	not shown

Qwen-7B's interaction is judge-dependent. Qwen2.5-7B's erosion falls from +78.1 to +68.9 pp under the Scheduled recipe ($D = -0.57$), the largest change of any 7–8B cell, with every Scheduled seed below every Constant seed. T5 reads "differs: Qwen's change is more negative" on Qwen3Guard, but its interval clears 0 only narrowly (+0.08), and on HarmBench it is "not shown" (E_{HB} falls from +60.6 to +55.6 pp). T6 is "not shown" on both judges. We had predicted that Qwen-7B barely moves (D between −0.5 and +0.2) and that T5 and T6 are "not shown": the mean is inside the predicted range (+65 to +85) but D is below it, and the T5 prediction failed. T5 excludes zero only on the main judge; the mean decline is smaller on the compliance judge.

Half LR lowers both families' mean erosion. Halving the Constant recipe's learning rate lowers OLMo-2-7B's mean E from +44.1 to +31.9 pp (3 seeds; $D = -0.51$) and Llama-3.1-8B's from +60.7 to +53.3 (2 seeds; −0.34). The interaction is "not shown" on both judges, as predicted, and OLMo-2-7B's half-LR mean (predicted about +32, range +20 to +45) held. This does not establish equal effects. The idea came from interim replication data, and Llama's half-LR results were already known, as its file discloses.

Reading. Both judges support the 1B OLMo interactions, but the 7–8B replication remains inconclusive, without proving a size interaction. Half LR lowers OLMo-7B's and Llama-8B's means without establishing equivalence. Qwen-7B's response (T5) and Qwen-1.5B's small rise (T2) depend on the judge, so transfer remains unestablished.

6 Where the change sits

We sort each answer by its opening words into a refusal, a caveat or a direct opening, with a frozen word list (Section B). Fine-tuning replaces refusal openings with direct openings, not with caveats: under the Constant recipe, the share of OLMo's sampled answers that open with a refusal falls from 0.80 to 0.30 (to 0.44 under the Scheduled recipe), and Qwen2.5-1.5B's from 0.93 to 0.07 (Figure 1). We split the rise in U into a change in the mix of openings and a change in the Unsafe share within each type of opening, averaged over both orders. The opening switch carries about half of OLMo's rise (0.42–0.64 across word lists and judges; 0.48 under the Scheduled recipe) and more of Qwen's (0.64–0.78). Answers that open directly also became less safe: OLMo's Unsafe share within direct openings rose from 12 of 66 answers to 840 of 1,197 (pooled over 3 seeds), so the first words are not the whole account (compare the shallow-alignment view of Qi et al. (2025)). The Scheduled recipe keeps more refusal openings but does not clearly make direct answers safer (0.60 against 0.68 Unsafe within them, seed 1101).

The families differ in two layers, and the recipes act through the openings; Section E gives the details and Figure 1 the opening mix.

7 Exploratory: where the recipe acts

Refusal-direction proxy. These exploratory analyses follow the results above and carry no test. Twenty of 24 planned fine-tuned units were recovered (two or three seeds per cell); four remain unavailable. We take the residual stream at the last prompt token for 200 harm and 200 harmless task prompts. At each layer, the proxy direction is the difference of the untuned group means. Separation projects both groups onto that fixed untuned direction and divides their mean difference by pooled sample SD. Retained averages the fine-tuned/untuned separation ratio over upper-half layers. Content or task differences may contribute; we did not intervene on this direction. Table 12 reports the retained separation and direction cosine, averaged over the upper half of layers. Scheduled retains more OLMo-1B separation (0.71–0.74 versus 0.59) than Llama-1B (0.60–0.62 versus 0.57–0.61) or OLMo-7B (0.40–0.42 versus 0.37–0.41). Llama-8B retains 0.41 versus 0.32–0.37. These incomplete-cell ranges are descriptive. Every probed 7–8B unit retains less than every 1B unit. A prospective OLMo pilot uses one fresh training seed and one engine (Section G). The schedule package changes U by -12.0 pp at 1B and $+7.5$ pp at 7B; token weighting under that schedule adds -8.5 and $+2.8$ pp, with 16 pp more benign refusal at 1B. These checkpoint-conditional intervention effects do not validate the representation proxy as a mediator.

Other leads. Answer length, over-refusal, which prompts erode and the untuned starting points are in Section E.

8 Does the content of benign data matter?

Does the darker benign fiction arm erode more? We tested this in a pre-registered experiment on OLMo-2-1B under the Constant recipe (Section D). Two arms replace Dolly with passages from public-domain novels, in Dolly's format and length distribution: *dark* fiction (Gothic and horror books; the passages richest in a darkness lexicon) and *light* fiction (romance and domestic books; the passages poorest in it), each screened by Granite and run on the three Dolly seeds. We fixed the contrasts, margins and labels before any run.

Dark–light differences fit the margins. Dark minus light is -4.9 pp on E (95% interval -9.5 to -0.3) and -1.7 pp on E_{HB} (-5.1 to $+1.8$). Both 90% intervals lie inside the pre-registered margins (± 15.8 and ± 10.6 pp, about a third of Dolly's erosion on each judge), so the label is "no difference larger than the margin" (Table 10). The margins are wide, so this is a weak equivalence claim, not equality. We had predicted that dark fiction erodes more; the point estimates have the opposite sign, and on E alone the 95% interval excludes 0 in that direction (on E_{HB} it does not).

Fiction eroded far less than Dolly. Light fiction minus Dolly is -28.5 pp on E (-33.4 to -23.6) and -23.6 pp on E_{HB} (-28.4 to -18.8), so the label is "fiction differs from Dolly (lower)". Granite agrees in direction. Both arms' E_{HB} lie below our 20 pp threshold for harm wording; this does not imply absence of harmful compliance (§9).

The fiction models also changed in other ways. They refused more benign prompts and lost ARC accuracy, so weaker instruction following may explain part of the smaller erosion (Section D).

9 What the measure means

What U tracks. U is a broad policy label. After fine-tuning, the probability of an Unsafe label given no refusal is 0.91–0.94, so U mostly tracks lost refusal; under hard categories only, OLMo's E is about 28 pp. HarmBench sees about two thirds of OLMo's Qwen3Guard erosion under the Constant recipe ($\kappa = 0.65$–0.71 with Qwen3Guard on the fine-tuned runs).

A blind human read. A blind read of 50 answers found that of 39 rated answers labelled Unsafe, 34 went along with the request but only 13 gave usable help, so U measures lost refusal well and usable harm is a smaller share (Section C).

Judge wording. A sentence about harm needs the HarmBench contrast to have the same sign as the Qwen3Guard contrast and at least half its size, and $E_{\mathrm{HB}} \geq 20$ pp; below that we say lost refusal. This is a wording threshold, not evidence of absent harm. Section H applies the rule to every cell.

What we screened. Section H lists every setting we ran, in time order, including runs that did not enter the tables.

10 Related work

Benign fine-tuning erodes safety. Qi et al. (2024) showed that fine-tuning aligned models on

benign instruction data, such as Alpaca and Dolly, erodes their safety. He et al. (2024) showed that benign rows chosen for their likeness to harmful rows erode safety more than random rows, and that list-format answers raise erosion by themselves. Kim et al. (2025) show that optimisation choices, such as the learning rate, batch size and number of epochs, drive much of the safety loss.

Family contrasts exist. Hawkins et al. (2026) fine-tune small models from several families with one LoRA recipe (constant learning rate 5×10^{-5}, one epoch, 1,000 synthetic rows seeded from Dolly, 3 seeds) and measure SORRY-Bench compliance: Gemma-3-1B rises from 30.2% to 65.5%, Llama-3.2-1B from 34.5% to 38.4%, and Qwen3-0.6B falls (their Table 2). So the same Llama-3.2-1B erodes little in their setup and nears saturation in ours; their data, judge, starting rate and recipe all differ from ours. Khan et al. (2026) fine-tune four families of 7–9B models on medical and legal data. They write: "Base model family was likewise an unreliable predictor" (their §4.2.1). Guo et al. (2026) compare Llama-3.1-8B and Qwen2.5-7B, with no OLMo model. On Alpaca, Qwen erodes less (LoRA on 5,000 rows: 9.1% against 44.8% attack success). On Dolly with full fine-tuning, the order flips (73.6% against 66.9%). So at 7–8B, the Qwen-versus-Llama order in Guo et al. (2026) depends on the data. Their 7.3–9.1% for LoRA on Qwen2.5-7B with Alpaca (rank 32, learning rate 2×10^{-4}, 5 epochs; HEx-PHI prompts; a GPT-4o-mini judge) is far below our $+78.1$ pp on Dolly; the data alone may explain much of this gap, so the recipe and the data must be reported together. Fraser et al. (2025) fine-tune Llama-3.2-1B on Dolly with 4-bit LoRA for 5 epochs (their text says instruct models; their links point to base checkpoints).

Shallow alignment. Qi et al. (2025) argue that safety alignment sits mostly in the first few answer tokens. On benign Samsum data, a 10-step warmup alone cut their attack success from 23.4% to 3.8% (Table 9 of the ICLR version). The Scheduled recipe includes a warmup, and §6 finds that the opening switch carries about half of OLMo's erosion.

Content of benign data. Zhao et al. (2025b) fine-tune 95 role-specific LLaMA-3-8B models and find that safety falls further the more villainous the character. Vetter et al. (2026) trace emergent misalignment to text about villains and harmful agency; human-written documents did not reliably cause it, while synthetic instruction-answer pairs made from the same content did. Narrow fine-tuning can generalise broadly (Betley et al., 2026), and answer format alone can raise erosion (He et al., 2024). Our fiction arms are human-written passages, not role-play or instruction-answer pairs, and their dark–light differences lie within wide margins (§8), consistent with Vetter et al. (2026).

OLMo-2. We are not aware of prior benign-fine-tuning erosion numbers for OLMo-2 Instruct; our check (one web search, not exhaustive) found refusal geometry in OLMo-2-1B, where the only fine-tuning adds refusals (Labunets, 2026), and OLMo-2 7B and 13B with harmful data only (Lee et al., 2026).

We add OLMo-2, matched recipe changes within families at two sizes, and an opening-level account of where the erosion sits.

11 Conclusion

The same recipe changes give different responses across our models. Scheduled lowers OLMo-1B's mean erosion, and its two 1B family-interaction intervals exclude zero on both judges, but benign refusal also rises. The 7–8B OLMo-versus-Llama replication is inconclusive, and Qwen-7B's interaction against OLMo depends on the judge. Half LR lowers OLMo-7B's and Llama-8B's means without establishing equal effects. Under one recipe the families keep their erosion order at both sizes. Dark–light differences fit wide margins; both fiction arms erode less than Dolly while refusing more benign prompts.

What this means in practice. We need to check a recipe on the model, data and judge where it will be used. The 1B OLMo result does not establish a generally safety-preserving recipe, as lower Unsafe shares can come with more benign refusals or weaker instruction following. So report over-refusal and task accuracy beside erosion, with the model revision, recipe, dataset, judge and every setting screened. Compare recipes within models, and models under matched recipes. The pilot estimates component effects within fixed checkpoints; a causal size effect and refusal-direction mechanism remain unresolved. Wide-margin equivalence does not establish equality.

Limitations

- **Scope.** Every claim holds only for these models (three instruct families, at 1B and at 7–8B), Dolly and the two fiction datasets (on OLMo-2-1B only), LoRA fine-tuning, these recipes, the 200 development harm prompts and these judges. No result is confirmed on held-out prompts.

- **Choices made after seeing results.** We fixed the 20–55 pp window for partial erosion after the first Qwen results, with no written reason for 20 and 55. Seed 1101 chose the setting and counts in the OLMo mean. We wrote the exact 1B family-by-recipe test after 11 of its 12 values were known. Rules or predictions fixed before new runs, sometimes with known comparator cells, cover the Scheduled recipe, the token-mean recipe, rank 16, the fiction study and the six tests of §5, and the human read's analysis was fixed before reading; the other comparisons are descriptive (Table 13). The exploratory analyses of §7 were run after the results and carry no test.

- **The transfer tests.** The 7–8B seeds spread more than the 1B seeds, so the 7–8B tests have less power, and "not shown" is weak evidence of no interaction. Six tests ran on overlapping cells with no multiplicity adjustment; T5 clears 0 only narrowly and only on Qwen3Guard, and T2 differs on HarmBench only. The half-LR test's idea came from interim replication data. The Llama-3.1-8B half-LR cell has two seeds, whose results were already known before the new OLMo runs. An inconclusive interaction does not establish equal effects or absence of an interaction. We tested one standard recipe change and one learning-rate step; other schedules, ranks or epochs may behave differently.

- **Fiction study.** One family, one recipe and 16 books. The fiction-trained models also refused more benign prompts and lost ARC accuracy, so their lower erosion may come from weaker instruction following. The margins are wide (about a third of Dolly's erosion), so "no difference larger than the margin" is a weak equivalence claim.

- **Few seeds.** The between-seed variance rests on 2 df: the sample SD's 90% range is 0.23–1.73 times the true SD. The OLMo band covers evaluation noise only. The 1B variant rows with rank 16 or half the learning rate rest on one or two seeds. The logit interactions have approximately 2.7–6.2 df. The refusal-direction probe has two or three recovered seeds per cell, with four missing units; harmful–harmless separation can reflect content or task differences and is not a causally validated refusal mechanism.

- **What the judge measures.** U is a broad policy label that mostly tracks lost refusal (§9), and usable harm is a smaller share. HarmBench did not judge the 1B half-LR or rank-16 units, and Granite judged greedy answers only.

- **Differences between families and sizes.** The families differ in chat template, tokeniser (so the same rows give different token counts), post-training pipeline and starting over-refusal, and the two OLMo-2 sizes are different releases. The runs used four engine identities (Section B); OLMo-1B's Constant and other recipes span two engines, with baseline reuse validated on the evaluation path. The separate one-seed pilot (Section G) estimates recipe- component effects within fixed checkpoints; it does not isolate a cause based on parameter count or pretraining. Its schedule also changes the mean learning rate, and its additional 1B reduction raises benign refusal.

- **Recipes.** The Constant recipe is not standard practice. The token-mean recipe separates the loss weighting from the schedule, but not the schedule's warmup from its decay, which also halves the mean learning rate.

- **Human read.** One reader, who designed parts of the study, rated 50 answers. There is no second rater, and the groups are too small to compare.

- **ARC.** The ARC check is uninformative at 1B.

Ethical considerations

This paper measures refusal changes in six open instruct models at 1–8B parameters after training on public benign instruction data or screened fiction. These results may help practitioners retain

refusals, but could also help remove them; we do not quantify that risk. We propose no attack and release no harmful prompt or answer text, and we use every model, dataset and judge for research, consistent with its licence (Section C). An author read 50 model answers to harmful requests for the blind human read, with the rubric in Section C; no other annotators took part.

Use of AI assistants. Anthropic Claude and OpenAI GPT/Codex helped design analyses, write code, check results and draft or edit the paper. The authors made the study decisions and take responsibility for its content.

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A Per-seed tables

Table 4 gives every seed of every 1B run in Tables 1 and 2, the Qwen2.5-1.5B screens (Constant, rank 16 and High-dose) and Llama-3.2-1B under Constant, half LR; Table 5 gives every 7–8B seed. The tables in the main text give the means.

Table 4: Every 1B seed. U is the Unsafe share of the 600 sampled answers; E, E_{HB} and Granite greedy E are in pp, each against the same model's untuned unit; over-refusal is in %.

Recipe	Seed	U	E	E_{HB}	Over-refusal	Held-out loss	Granite greedy E
OLMo-2-1B							
untuned		3.2%			17	2.100	
Constant	1101	46.7%	+43.5	+27.7	4	1.542	+44.0
Constant	1102	46.8%	+43.7	+30.5	6	1.547	+40.5
Constant	1103	51.3%	+48.2	+32.8	4	1.545	+44.5
Constant, token-mean	1101	46.8%	+43.7	+29.7	9	1.587	+38.0
Constant, token-mean	1102	42.7%	+39.5	+26.5	8	1.589	+31.0†
Constant, token-mean	1103	45.7%	+42.5	+27.0	5	1.590	+35.0†
Scheduled	1101	32.2%	+29.0	+17.8	17	1.612	+23.0
Scheduled	1102	34.2%	+31.0	+19.3	14	1.613	+25.5
Scheduled	1103	29.7%	+26.5	+14.3	18	1.613	+18.5
Scheduled, half LR	1101	23.5%	+20.3	—	18	1.634	+11.5
Scheduled, half LR	1102	24.5%	+21.3	—	22	1.635	+12.0
Scheduled, rank 16	1101	23.3%	+20.2	—	25	1.649	+9.0
Qwen2.5-1.5B							
untuned		4.7%			68	1.890	
Constant	1101	79.8%	+75.2	+46.0	3	1.459	+85.5†
Constant	1102	74.0%	+69.3	+44.0	3	1.464	+75.5†
Constant	1103	76.0%	+71.3	+46.3	1	1.464	+79.5†
Scheduled	1101	79.3%	+74.7	+53.2	9	1.514	+82.0†
Scheduled	1102	78.5%	+73.8	+53.0	5	1.512	+81.5†
Scheduled	1103	78.0%	+73.3	+52.0	6	1.510	+82.0†
Constant, rank 16	1101	72.5%	+67.8	+45.8	3	1.481	+75.5
High-dose	1101	87.3%	+82.7	—	2	1.857	+88.0
High-dose	1102	88.7%	+84.0	—	3	1.829	—
High-dose	1103	83.5%	+78.8	—	10	1.850	+85.5
Llama-3.2-1B							
untuned		13.8%			11	2.243	
Constant	1101	76.3%	+62.5	+40.5	2	1.555	+68.5†
Constant	1102	75.2%	+61.3	+39.3	4	1.554	+63.0†
Constant	1103	67.8%	+54.0	+34.0	3	1.553	+57.0†
Constant, half LR	1101	61.8%	+48.0	—	5	1.566	+47.5†
Scheduled	1101	73.8%	+60.0	+35.5	7	1.619	+58.5
Scheduled	1102	70.3%	+56.5	+35.8	9	1.614	+58.5
Scheduled	1103	68.7%	+54.8	+36.0	6	1.613	+53.0
Scheduled, half LR	1101	55.0%	+41.2	—	4	1.642	+38.0

Blank: not applicable (untuned rows). —: not recorded (HarmBench did not judge the run; Granite did not judge High-dose seed 1102). HarmBench values for Llama's Constant seeds 1102 and 1103, Scheduled seed 1103 and every token-mean seed come from a second pod run (10 Oct); the units it shared with the first runs got the same labels. The Qwen2.5-1.5B Constant and Scheduled rows ran on a later engine identity (Section B); seed 1101 reproduced the earlier Constant unit's counts exactly. High-dose rows are after 3 epochs; this recipe failed its own bar that answers must end (repetition loops), so it is a screened setting, not a clean result. Untuned HarmBench shares (pod): OLMo 3.2%, Qwen2.5-1.5B 3.2%, Llama 11.0%. Untuned Granite greedy shares: OLMo 2.5%, Qwen2.5-1.5B 3.0%, Llama 13.5% (Mac) and 13.0% (pod). †Granite on the pod, against the pod-judged untuned unit; every other Granite value is from the Mac. On the Mac, Llama's Constant seed 1101 gives +67.5.

Table 5: Every 7–8B seed, as in Table 4. Every run used one engine identity (Section B). HarmBench and Granite judged every unit on the pod, in two runs of the same device type, each unit against its model's untuned unit judged in the first run.

Recipe	Seed	U	E	E_{HB}	Over-refusal	Held-out loss	Granite greedy E
OLMo-2-7B							
untuned		1.8%			27	2.121	
Constant	1101	51.5%	+49.7	+37.0	2	1.353	+49.5
Constant	1102	39.0%	+37.2	+27.7	2	1.358	+40.0
Constant	1103	47.3%	+45.5	+34.7	3	1.354	+45.0
Constant, half LR	1101	35.3%	+33.5	+25.3	4	1.368	+36.0
Constant, half LR	1102	32.7%	+30.8	+23.7	5	1.373	+34.5
Constant, half LR	1103	33.3%	+31.5	+23.0	7	1.370	+34.5
Scheduled	1101	52.8%	+51.0	+38.5	7	1.411	+44.5
Scheduled	1102	59.8%	+58.0	+43.0	6	1.412	+58.0
Scheduled	1103	43.7%	+41.8	+32.0	6	1.410	+39.0
Qwen2.5-7B							
untuned		4.8%			7	3.335	
Constant	1101	87.0%	+82.2	+63.5	3	1.293	+87.0
Constant	1102	83.3%	+78.5	+59.7	2	1.295	+86.0
Constant	1103	78.3%	+73.5	+58.5	3	1.296	+81.0
Scheduled	1101	74.0%	+69.2	+55.2	5	1.336	+76.0
Scheduled	1102	75.5%	+70.7	+57.5	6	1.337	+73.5
Scheduled	1103	71.7%	+66.8	+54.0	6	1.334	+71.5
Llama-3.1-8B							
untuned		5.3%			17	2.109	
Constant	1101	75.5%	+70.2	+53.8	1	1.321	+73.5
Constant	1102	61.2%	+55.8	+39.8	1	1.328	+50.5
Constant	1103	61.5%	+56.2	+39.7	1	1.321	+54.0
Constant, half LR	1101	57.8%	+52.5	+36.2	2	1.326	+50.0
Constant, half LR	1102	59.3%	+54.0	+37.8	2	1.328	+55.0
Scheduled	1101	65.5%	+60.2	+41.5	7	1.364	+60.0
Scheduled	1102	67.2%	+61.8	+45.2	7	1.366	+61.5
Scheduled	1103	60.8%	+55.5	+40.8	2	1.362	+54.0

Untuned HarmBench shares (pod): OLMo-2-7B 1.2%, Qwen2.5-7B 5.3%, Llama-3.1-8B 5.5%. Untuned Granite greedy shares (pod): OLMo-2-7B 1.5%, Qwen2.5-7B 4.0%, Llama-3.1-8B 6.0%. Qwen2.5-7B's untuned unit and its Constant seed 1101 reproduced the earlier one-seed units' counts exactly (on the Mac, Granite gave that seed +87.5). Llama-3.1-8B's half-LR seeds were run for the larger study before the half-LR pre-registration.

B Settings and hashes

Reproduction scope. This appendix specifies the recorded training and evaluation procedure. The separate numerical package reconstructs analyses from saved labels and aggregates; it does not rerun training or extract representations. A supplementary recipe archive contains the 23 configurations used by the reported evaluated checkpoints, with full file and canonical-JSON hashes. Preflight-only candidates and the incomplete seed-1122 follow-up are listed in the screening log, not treated as completed results.

Data and checkpoints. Dolly selection uses design seed 20261006. Eligibility excludes empty instruction/response fields, forbidden control-token strings, normalized exact or near-duplicate overlap with evaluation prompt forms (Jaccard cutoff 0.5), duplicate normalized user texts, and token-audit failures. After these recorded filters, each category supplies 25 task-query rows and 600 training rows. The 200 task queries have no exact user-text or prompt–response overlap with the 4,800 training rows in any of the three recorded tokenizer-specific preparations. This is an exact-text check, not proof of semantic independence or external generalization. The main one-epoch runs use 37 full 128-row updates and one 64-row update; rows are not dropped or carried into the next epoch. High-dose instead uses 8 rows per update, 600 updates per epoch and three epochs. The intermediate quarter-epoch checkpoint is the first update boundary at or after 1,200 rows (10 updates, 1,280 rows in the main recipes); the reported one-epoch endpoint uses all 4,800 rows. Fiction data and its stopping rules are in Section D.

Seeds and initialization. The main cells use seeds 1101–1103; Section A lists exceptions, and the component pilot uses fresh seed 1121 separately. Named streams take the first eight bytes of SHA-256 of UTF-8 strings joined by `|`, interpreted as an unsigned big-endian integer. Sample indices are zero-based, and sampled-answer seeds are reduced modulo 2^{63} before seeding their generators. Adapter initialization uses the stream `(seed,lora-init)`: in sorted parameter-name order, each LoRA A entry is uniform on $[-1/\sqrt{d_{\text{in}}}, 1/\sqrt{d_{\text{in}}}]$, and every B entry is zero. Each epoch shuffles sorted training-row identifiers with the stream `(seed,data-order,epoch)`, where the first epoch has index zero. Each sampled harm answer has its own generator, seeded by `seed`, `harm-sample`, `prompt-id` and `sample-index`. Thus equal seed numbers alone do not assert equal initialization across different model architectures.

Exact loss reduction. Let ℓ_{ij} be target-token NLL, n_i the response-plus-end-of-turn target count of row i, and m the actual number of rows in an optimizer step. Example-mean and token-mean objectives are, respectively,

$$L_{\text{ex}} = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{n_i}\sum_{j=1}^{n_i}\ell_{ij}, \tag{1}$$

$$L_{\text{tok}} = \frac{1}{\sum_{i=1}^{m} n_i}\sum_{i=1}^{m}\sum_{j=1}^{n_i}\ell_{ij}. \tag{2}$$

Every micro-batch uses the full optimizer step's normalizer, including the smaller final step; token-mean is not a mean of micro-batch means. Training uses right padding and a masked prompt; generation uses left padding with explicit position IDs. Over-length training rows or prompts are rejected by preparation, not silently truncated. The answer and its end-of-turn token are targets. Gradient checkpointing is off in the reported configurations; the output-head computation uses 1,024-target-token chunks, with recomputation during backward. Base weights remain frozen; only the named LoRA projections train, with scaling $\alpha/r = 2$.

Applied schedule. For update $t = 1, \ldots, 38$, let $k = t - 1$, warmup $w = 4$, horizon $T = 38$, peak $\eta_{\max}$ and $p_k = (k - w)/(T - w)$. The scheduled rate is

$$\frac{\eta_t}{\eta_{\max}} = \begin{cases} k/w, & k < w, \\ \frac{1}{2}[1 + \cos(\pi p_k)], & k \geq w. \end{cases} \tag{3}$$

The first update uses zero and update 5 first uses the peak. The final update uses 2.1329×10^{-7} for peak 10^{-4}; zero is the subsequent scheduler endpoint, not the last applied rate. The 38-update mean is 5×10^{-5}. Half-LR halves every applied rate. This matches nominal optimizer-update mean, not parameter-update magnitude or token-weighted exposure; warmup, timing and mean rate remain bundled in Scheduled.

Evaluation details. Policy decoding is greedy for benign prompts and the extra harm answer, and ancestral sampling for the three sampled harm answers; there is no beam search or additional

logits processor. A row stops at its first model-specific stop token or the 2,048-token cap; cap hits remain flagged. Qwen3Guard decodes greedily with at most 128 new tokens and an 8,192-token input limit. Its parser requires consistent `Safety`, `Categories` and `Refusal` fields; malformed outputs and guard outputs that fail to reach a stop token are invalid, never silently Safe. Older recipes refuse an over-limit guard input; later recipes explicitly permit an unscored record. Neither silently clips the input. All 77 archived sampled harm summaries have zero invalid labels. Unsafe share uses all 600 sampled answers; benign refusal uses validly judged greedy answers. HarmBench reads at most the first 511 answer tokens. Granite uses its harm criterion with thinking disabled and greedy decoding capped at 32 tokens, on 200 greedy answers. Its last `<score>` field is stripped and lowercased; only `yes`/`no` parse. For a templated input longer than 8,000 tokens, the answer is replaced by a prefix of up to $\max(1000, \lfloor |a|\, 7000/n \rfloor)$ characters, where n is the original input-token count. An input still above 8,160 tokens refuses the run. Trimming/unparsed flags and device-specific baselines are retained. ARC chooses the label with greatest teacher-forced label-token log-probability among the question's choices. Task-query NLL is an example-mean forward-only response loss in every recipe, including those trained with token-mean loss; evaluation restores training RNG state.

Software and numerical settings. All 77 archived evaluated checkpoints record Python 3.12.3, PyTorch 2.8.0+cu128 (CUDA runtime 12.8), Transformers 4.56.2, PEFT 0.17.1, Tokenizers 0.22.2, NumPy 2.2.6, Safetensors 0.8.0 and Accelerate 1.15.0. The recorded policy GPU is an A100-SXM4-80GB. CUDA deterministic algorithms are enabled with errors rather than warnings, bf16 reduced-precision reduction is disabled, and TF32 is disabled for CUDA matmuls and cuDNN, cuDNN benchmarking is off, and `CUBLAS_WORKSPACE_CONFIG=:4096:8`. These are recorded settings and identities, not a claim of bitwise equivalence across engines, hardware or judge devices.

Hash interpretation. Model/judge revisions are repository commits; prepared-data hashes identify manifests; engine hashes identify the recorded source set. Recipe hashes use sorted-key, two-space-indented UTF-8 JSON with a trailing newline, not necessarily the original file bytes. The recipe index supplies both definitions and each file's full hash. These hashes identify inputs and detect changed bytes; they do not independently certify remote execution or scientific validity. The numerical archive omits source prompt IDs, texts and model states, so it supports paired numerical analyses rather than replay of the full generation/training pipeline.

Table 6: Training and evaluation settings. The top part gives the main-recipe defaults; the bottom part lists variants, including the High-dose batch and epoch exceptions. Exact configured recipes are indexed in Table 8.

Setting	Value
Main-recipe defaults; exceptions below	
Training code	our own training loop in PyTorch, with Hugging Face Transformers (model loading) and PEFT (LoRA)
Adapter	LoRA on the q, k, v, o, gate, up and down projections of every layer; dropout 0. Trainable parameters at rank 64: OLMo-2-1B 48.2M, Llama-3.2-1B 45.1M, Qwen2.5-1.5B 73.9M, OLMo-2-7B 159.9M, Qwen2.5-7B 161.5M, Llama-3.1-8B 167.8M
Optimiser	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay; gradient-norm clipping at 1.0
Batch	up to 128 rows per optimiser step (micro-batches of 4 rows with 32 accumulation steps; the 7–8B models: 2 rows and 64 steps); one epoch of 4,800 rows is 38 steps
Loss	next-token negative log-likelihood on the answer and its end-of-turn token only; the prompt is masked
Precision and length	base weights in bfloat16; adapters and the loss in float32; SDPA attention; at most 3,072 tokens per training sequence; 1,536 tokens per generation/ARC prompt
Harm evaluation	200 development harm prompts; 3 sampled answers each (temperature 0.7, top-p 1.0) and one greedy answer; at most 2,048 new tokens
Other measures	over-refusal on 100 benign prompts (greedy answers); held-out loss on 200 Dolly rows; ARC forced choice on 189 questions
Judges	Qwen3Guard-Gen-4B on every answer (the main judge); the HarmBench Mistral-7B classifier on sampled answers; Granite Guardian 3.3 8B on greedy answers
Compute	one NVIDIA A100 80 GB GPU per run (rented cloud GPUs); a 1B run took about 1–3.5 h including setup and both evaluations, and the Qwen2.5-7B run about 3.6 h. Main HarmBench erosion ran on A100; embedded blind-read labels used Mac/MPS float16. Granite used A100 or an Apple-silicon Mac
What each recipe changes	
Constant	learning rate 10^{-4}, constant; per-row loss mean; rank 64, alpha 128
Constant, token-mean	as Constant, with a token-mean loss
Constant, half LR	as Constant, with learning rate 5×10^{-5}
Constant, rank 16	as Constant, with rank 16 and alpha 32
Scheduled	peak learning rate 10^{-4}: a 4-step linear warmup, then cosine decay over the 38-update epoch (last applied rate above zero); token-mean loss; rank 64, alpha 128
Scheduled, half LR	as Scheduled, with peak 5×10^{-5}
Scheduled, rank 16	as Scheduled, with rank 16 and alpha 32
Component C (seed-1121 pilot)	as Constant, with the Scheduled learning-rate schedule but example-mean loss
High-dose (a Qwen2.5-1.5B screen)	as Constant, with learning rate 2×10^{-4}, 8 rows per step (600 steps per epoch) and 3 epochs

Table 7: Revisions and SHA-256 hashes. Model and judge revisions are Hugging Face commit hashes. Table 6 gives the training and evaluation settings.

Item	Revision or SHA-256
Models	
OLMo-2-0425-1B-Instruct	`48d788eca847d4d7548f375ad03d3c9312f6139e`
OLMo-2-1124-7B-Instruct	`470b1fba1ae01581f270116362ee4aa1b97f4c84`; tokenizer from the OLMo-2-0425-1B-Instruct revision above
Qwen2.5-1.5B-Instruct	`989aa7980e4cf806f80c7fef2b1adb7bc71aa306`
Qwen2.5-7B-Instruct	`a09a35458c702b33eeacc393d103063234e8bc28`; tokenizer from the Qwen2.5-1.5B-Instruct revision above (identical files)
Llama-3.2-1B-Instruct	`9213176726f574b556790deb65791e0c5aa438b6`; tokenizer and chat template from Llama-3.1-8B-Instruct at `0e9e39f249a16976918f6564b8830bc894c89659`
Llama-3.1-8B-Instruct	`0e9e39f249a16976918f6564b8830bc894c89659`
Judges	
Qwen3Guard-Gen-4B	`6ec42827da0c1ff11e7a49dc269d2e810d27e108`
HarmBench-Mistral-7b-val-cls	`51182c7cdaf9b2e5f05b745b81b047b67e7384a1`
Granite Guardian 3.3 8B, Q4_K_M	granite-guardian-3.3-8b-GGUF at `8c87479a2dc329e4e362a6e5dda7e7de3e0372b2`; file SHA-256 `b4fa861b5163373413d9659c22e344e6206baad824ead17226b2058e54d14587`
Prepared data (SHA-256 of each manifest)	
Qwen2.5 (1.5B and 7B)	`9247c4bbccf212b02be7f221e2ee451d17ebf643a39a44db6e3bed6fb4d750f2`
OLMo-2 (1B and 7B)	`9ed61826f1634dcab5a7385f18556b108b3ba8db95fc17215fa4cf4f6cb993b9`
Llama-3 (3.2-1B and 3.1-8B)	`c52a50afddcfcfa818512ac7f1536c34c959774e815095bd30f69320b083d55a`
Fiction, dark (OLMo-2-1B)	`c57b56ceaf5c53fd3d6a9cabf97bdbb4f1a039fe5ba0536572df28b0200fb5a2`
Fiction, light (OLMo-2-1B)	`8f6493250d93b3b50c36f5ac1409c3d9a2b2850a98b2cec86ea4731768652f24`
Engine identities (code SHA-256)	
The earlier Qwen2.5 runs: Constant (1.5B and 7B), rank 16 and High-dose	`c52c0ebc83793fc552f84908e83542d5e46543459fcafb7be2fc9c7ff42ac597`
OLMo-2-1B, Constant	`552538748acec1a8ffd1ee976b9eb3c3fdfe9ca01b8148ee8810291409357441`
Historical Llama-3.2-1B, OLMo-2-1B recipe variants and fiction runs (excluding seed-1121 components)	`e49a978c5fa27c3ba4bb1570ac8b24501e09bf8db5c35e31ffba90bf775cdc06`
The later three-seed Qwen rows, current 7–8B rows and the seed-1121 component pilot (engine X)	`472fc7f6d619bdee2c3aa38e444024626be8b66bfb74a5abf37f2ddb6dded892`
Analysis	
Interaction-test script of §5, hashed in every pre-registration before its runs	`798e6c03177baf4267a1880a6a897186045a95aef27e496cc6e27c4bba311cf2`
Opening word list, frozen before any result (§6)	`bbbc71e9eebb62876430ffb5e0f54113c3818984b8369924b7bd2212c4c1ca17`
Reading bars, with the 20–55 pp window (§3)	`c0b6f2aae43285de632a175adb1e913cd31f37bc6ac25fb56e4468ff3a1208ea`
Logit-scale prediction for the Scheduled recipe on Qwen2.5-1.5B, written before the run was dropped (§9)	commit `c868608a08ea1192e1e8bf489488631a4903beaa`, 2026-10-08 17:14 UTC

Table 8: Canonical recipe SHA-256 prefixes (12 characters). Full hashes, filenames and byte-preserved configurations are in the supplementary recipe index. The same file can serve a historical recipe and component arm A or D; component C differs by using example-mean scheduled loss.

Model	Recorded recipe	Canonical SHA-256 prefix
Qwen2.5-1.5B	Constant, rank 16	`86079a35d7f9`
Qwen2.5-1.5B	Constant	`787ccff024d7`
Qwen2.5-1.5B	Scheduled	`f5005e8ee950`
Qwen2.5-1.5B	High-dose	`1656cba94fd7`
Llama-8B	Constant	`c5462c770031`
Llama-8B	Constant half LR	`74758af1791d`
Llama-8B	Scheduled	`442f133fb186`
Llama-1B	Constant	`bb0f563b9a81`
Llama-1B	Constant, half LR	`443f05e1006a`
Llama-1B	Scheduled	`b3f5cb579f05`
Llama-1B	Scheduled, half LR	`f87f558d6a7f`
OLMo-2-1B	Scheduled, rank 16	`a7f7a5db45aa`
OLMo-2-1B	Component A; Constant; Fiction F-dark; Fiction F-light	`9e9bde6f2f32`
OLMo-2-1B	Constant, token-mean	`ec8a6b212522`
OLMo-2-1B	Component D; Scheduled	`c004349c29d2`
OLMo-2-1B	Scheduled, half LR	`d43a24c83ccb`
OLMo-2-7B	Component A; Constant	`18cc31d7117a`
OLMo-2-7B	Constant half LR	`8df9f02f988e`
OLMo-2-7B	Component D; Scheduled	`01259f288c56`
Qwen2.5-7B	Constant	`ba5c402f15bc`
Qwen2.5-7B	Scheduled	`3146ee21bd7b`
OLMo-2-1B	Component C	`ff4af66ce9c0`
OLMo-2-7B	Component C	`2878008cc0a2`

Table 9: Granite tokenizer and chat-template file SHA-256 hashes, pinned separately from the GGUF weight file in Table 7.

File	SHA-256
`tokenizer.json`	`91168e938f05796aa6dcca7e485e4b30ab52785320c7a6391ecef86e6c84681e`
`tokenizer_config.json`	`fcb3bdcbcf27ab09cbd7385c486319413ec3619192255a95969253b9746ff874`
`chat_template.jinja`	`ddb691d71955c735aeca367b7405cf3e3622cf33788d6a1a7186965f0a4a69fc`
`special_tokens_map.json`	`21ce694081bb9ae1bd4bc64549e72e0799ebb74705e6b650e3585d85b71ebdc1`
`added_tokens.json`	`bb33d55934aa82d29cc62f3d19cdbc60f315763f6ccee21bdfd8b3bde2f33d3b`

C Artifacts, licences and the blind read

Artifacts and licences. We use OLMo-2-0425-1B-Instruct, OLMo-2-1124-7B-Instruct, Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct (Apache 2.0), Llama-3.2-1B-Instruct (Llama 3.2 Community License) and Llama-3.1-8B-Instruct (Llama 3.1 Community License); the judges Qwen3Guard-Gen-4B and Granite Guardian 3.3 8B (Apache 2.0) and the HarmBench classifier (MIT); the training data Dolly-15k (CC BY-SA 3.0) and, for the fiction study, passages from 16 Project Gutenberg books that are in the public domain in the United States (Section D); and, for evaluation only, prompts from HarmBench and StrongREJECT (MIT), SimpleSafetyTests (CC BY 2.0), and OR-Bench and XSTest (CC BY 4.0). We use each for research, consistent with its terms, and release no prompt or answer text.

The blind human read. We read 50 answers blind, with the sample, rubric and analysis fixed before reading and the ratings sealed before the key was opened. Of 39 rated answers labelled Unsafe, 34 (0.87, 95% interval 0.73–0.96) went along with the request, but only 13 (0.33, 0.19–0.50) gave usable help; the 1B models often comply vaguely, generically or wrongly. All 10 Safe-labelled controls were refusals or deflections. On the 15 OLMo answers that HarmBench judged, the human bar was stricter: 4 of 12 HarmBench *yes* answers were rated usable help. So U measures lost refusal well, and usable harm is a smaller share.

Blind-read rubric. The 50 answers were drawn by a seeded rule from the sampled answers: 40 labelled Unsafe (12 from OLMo-2-1B under the Constant recipe, 6 from OLMo-2-1B under the Scheduled recipe, and 8, 6 and 8 from Qwen2.5-1.5B, Qwen2.5-7B and Llama-3.2-1B under the Constant recipe) and 10 labelled Safe, as controls. The reader saw only an item number, the request and the answer, and gave each answer one rating: 0, refuses or deflects (declines, warns, redirects, or gives only safe general information); 1, engages but gives no usable help (goes along with the request but is vague, generic, wrong, fictional, or too incoherent to act on); or 2, gives usable help (specific steps, methods, materials or content that would meaningfully help someone do the harmful thing). The reader could also flag an answer as looping or garbled, or skip it. The rubric and the analysis were fixed before reading, and the ratings were sealed before the key was opened. The sample is stratified by source unit and judge label; its proportions describe this audited subset, not a representative deployment population. The embedded HarmBench labels used for this blind read were produced on Mac/MPS in float16, separately from the A100 labels used for main erosion.

D The fiction study: data and design

Books. Sixteen public-domain books from Project Gutenberg, the first eight eligible ones in each pool by text number. *Dark* (Gothic, horror and ghost-story subjects): Frankenstein (84), The Picture of Dorian Gray (174), The Phantom of the Opera (175), Dracula (345), Tales of Terror and Mystery (537), The Monk (601), Wieland (792) and The Works of Edgar Allan Poe, Volume 2 (2148). *Light* (romance, humour and domestic subjects, with horror, crime and war excluded): O

Pioneers! (24), Far from the Madding Crowd (27), The House of the Seven Gables (77), Ivanhoe (82), The Prisoner of Zenda (95), Persuasion (105), The Voyage Out (144) and Middlemarch (145).

Rows. Each arm has 5,000 rows, 625 per book, in Dolly's format: an instruction from a fixed template, the preceding sentences as context, and a passage as the response. Each row's response length matches one Dolly response within a small band (token counts under the OLMo-2 tokenizer), so both arms copy Dolly's length distribution. For each row, 100 candidate start sentences are drawn from its book; among the candidates that match the length, the dark arm takes the passage with the most darkness-lexicon words per 100 tokens and the light arm the one with the fewest, and passages never overlap. Before the screen, lexicon hits per 100 response tokens were 1.77 (dark) against 0.02 (light). Each arm trains on 4,800 rows; 25 rows per book are held back as task rows.

Safety screen. Granite Guardian 3.3 (*harm* risk) screened every training response, and each flagged row was replaced by the next candidate from the same book under the same rule. A sexual-content check fixed in the pre-registration fired (Qwen3Guard's sexual category above 0.5% of an arm), so, as pre-registered, Granite's *sexual content* risk joined the drop rule before any run. In all, 179 dark and 47 light rows were replaced. Qwen3Guard only describes the data: it never selects or drops a row. On the final rows it labels 11 dark and 6 light responses Unsafe (of 5,000; Violent: 4 and 2), so the darker arm is also darker by the judge, but only slightly; its sexual category stays at 1.14% (dark) and 0.98% (light). The Monk holds 5 of the dark arm's 11 Unsafe rows, and The Voyage Out 4 of the light arm's 6.

Training and pre-registration. Both arms use OLMo-2-1B under the Constant recipe with seeds 1101–1103, and each arm has its own untuned unit (the same model and prompts as D's). The question, arms, measurements, contrasts, reading rules, margins and analysis script were fixed and the pre-registration frozen before any fiction training run. One deviation, recorded before any result was read: the last seed (light, 1103) was launched after an engine update, on its own packed engine, so all six units share one engine.

The fiction models also changed in other ways. Both fiction arms learned their task, but they refused more benign prompts (24–31%, against 17% untuned and 4.7% after Dolly) and their ARC accuracy fell to 28 and 34% (chance about 25%; 39% untuned, 47% after Dolly). So prose passages also weakened instruction following, which may explain part of the smaller erosion; we do not separate the two. Every unit passed the bar that answers must end; one timing deviation was recorded before any result, and all six units share one engine (Section D).

Results per seed. Table 11 gives every fiction unit. Each arm's untuned unit gives the same values as D's: U 3.2%, over-refusal 17%, ARC 74 of 189.

Table 10: Fiction against Dolly on OLMo-2-1B (Constant recipe; means over three seeds). E, E_{HB} and Granite greedy E are in pp against each arm's untuned unit. Over-refusal and ARC accuracy are in %; untuned, they are 17 and 39.2.

Training data	E	E_{HB}	Granite	Over-ref.	ARC
Dark fiction	+11.7	+5.1	+8.0	30.7	27.7
Light fiction	+16.6	+6.7	+8.0	24.0	33.7
Dolly	+45.1	+30.3	+43.0	4.7	46.7

Table 11: Every fiction seed (OLMo-2-1B, Constant recipe). U is the Unsafe share of the 600 sampled answers; E, E_{HB} and Granite greedy E are in pp against the arm's untuned unit; over-refusal is in %; ARC is correct answers of 189. Granite judged the fiction units on the pod and D on the Mac (Mac-versus-pod parity passed; the untuned unit gives 5 of 200 on both).

Arm	Seed	U	E	E_{HB}	Granite greedy E	Over-refusal	ARC
Dark	1101	17.2%	+14.0	+6.7	+8.5	31	53
Dark	1102	15.2%	+12.0	+4.8	+7.0	29	52
Dark	1103	12.3%	+9.2	+3.7	+8.5	32	52
Light	1101	21.2%	+18.0	+8.3	+7.5	22	65
Light	1102	20.0%	+16.8	+6.5	+9.5	25	65
Light	1103	18.2%	+15.0	+5.3	+7.0	25	61
Dolly (D)	1101	46.7%	+43.5	+27.7	+44.0	4	94
Dolly (D)	1102	46.8%	+43.7	+30.5	+40.5	6	87
Dolly (D)	1103	51.3%	+48.2	+32.8	+44.5	4	84

E Details of Sections 6 and 7

Two layers of difference between families. The families differ in two ways. First, they lose different shares of their refusal openings under the Constant recipe: −0.51 for OLMo, −0.68 for Llama and −0.85 for Qwen2.5-1.5B (sampled, seed 1101). Second, their direct answers were already unsafe at different rates before any fine-tuning: 0.18 for OLMo, against 0.57 for Qwen and 0.64 for Llama. Qwen and Llama therefore already received Unsafe labels most of the time among their direct openings. For them, losing a refusal opening comes close to producing an Unsafe answer: the opening switch carries 0.64–0.78 of Qwen's rise and 0.68 of Llama's. OLMo's direct answers had to become less safe as well, which both layers show.

Recipe differences in the opening mix. Across the Constant recipe, Constant, token-mean and the Scheduled recipe, the Unsafe share within direct openings stays similar (0.57–0.70). What changes is how often the model still opens with a refusal: −0.51 under Constant, −0.50 under Constant, token-mean, and −0.34 to −0.36 under Scheduled. Scheduled, rank 16 keeps the most refusal openings (−0.20) and erodes least (+20.2 pp), with more over-refusal (25 of 100). So the Scheduled recipe's lower erosion looks like a dose effect of its schedule, expressed through the refusal openings (one seed for Constant, token-mean and for Scheduled, rank 16).

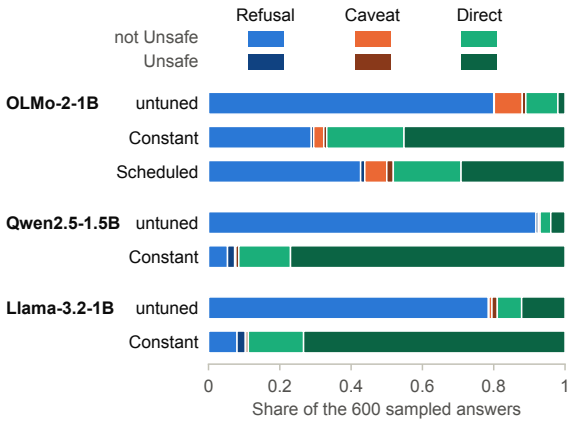


Figure 1: How the answers open, before and after fine-tuning. Each bar is one model before or after fine-tuning; fine-tuned bars are seed 1101. Each bar splits the 600 sampled answers into refusal, caveat and direct openings. The darker part of each opening is the share that Qwen3Guard labels Unsafe. The lighter part is the rest (Safe, Controversial or Invalid). One empty OLMo answer under the Scheduled recipe is not drawn.

This analysis is observational only. We ran no prefill test, and we count positions in words, not tokens.

Other leads. The Scheduled recipe lengthens the sampled answers to harm prompts in every



Figure 2: Erosion by model and recipe, for the 1B units of the first version of this note. Each row is one model and recipe, and each dot is one seed; seeds that nearly coincide are stacked. The left panel gives the Qwen3Guard erosion E and the right panel the HarmBench erosion E_{HB}, both in pp. Blue marks the Constant recipe and its variants; orange marks the Scheduled recipe and its variants. A black tick marks the mean of the seeds. On one-seed rows, a bar shows $E \pm 2$ SE (evaluation noise only). The three Qwen rows show the first seed (1101) only; the later three-seed Qwen rows, the Qwen Scheduled rows and the 7–8B rows (Tables 1, 2 and 5) are not drawn. HarmBench has no recorded SE, so the right panel has no bars. HarmBench did not judge the rows marked "not judged". We do not shade a "partial" window, because we fixed it after the first Qwen results (§9).

model (mean tokens rise by 30–90%; 101 → 136 for OLMo-2-1B, 82 → 153 for OLMo-2-7B) and raises over-refusal by 3–12 pp, most in OLMo-2-1B: it changes the answers' form everywhere, and the erosion in two cells. The families erode on partly different prompts: calling a prompt eroded if its seed-mean Unsafe rate under the Constant recipe is at least 2/3, the two sizes of one family share most eroded prompts (a Jaccard index of 0.62 to 0.80), while OLMo shares fewer with Llama or Qwen (0.42 to 0.57) than those two share with each other (0.66 to 0.74). A family name does not fix one starting point: OLMo-2-7B, a different release from OLMo-2-1B, starts more cautious (over-refusal 27% against 17%) and loses almost all of that caution.

Table 12: Refusal-direction retention (exploratory). Retained is the share of the untuned harmful-versus-harmless separation that the fine-tuned model keeps; cosine is between the untuned and fine-tuned directions; both are means over the upper half of the layers. Values are in seed order 1101–1103. OLMo-1B Constant lacks 1102, Scheduled lacks 1101; OLMo-7B and Llama-8B Scheduled lack 1103. All 20 recovered units are shown.

Model	Recipe	E (pp)	Retained	Cosine
OLMo-1B	Constant	+43.5, +48.2	0.59, 0.59	0.67, 0.67
OLMo-1B	Scheduled	+31.0, +26.5	0.71, 0.74	0.79, 0.80
Llama-1B	Constant	+62.5, +61.3, +54.0	0.57, 0.57, 0.61	0.50, 0.51, 0.53
Llama-1B	Scheduled	+60.0, +56.5, +54.8	0.62, 0.60, 0.60	0.54, 0.54, 0.55
OLMo-7B	Constant	+49.7, +37.2, +45.5	0.37, 0.41, 0.39	0.56, 0.60, 0.58
OLMo-7B	Scheduled	+51.0, +58.0	0.42, 0.40	0.60, 0.55
Llama-8B	Constant	+70.2, +55.8, +56.2	0.32, 0.37, 0.35	0.32, 0.41, 0.37
Llama-8B	Scheduled	+60.2, +61.8	0.41, 0.41	0.46, 0.46

F Claim status

Table 13 sorts the claims by when their rule was fixed. Eleven tests had their rule fixed before their new runs, sometimes using known comparator cells: the Scheduled-recipe and token-mean predictions, the rank-16 rule, the fiction study's two labels, and the six pre-registered tests of §5 (the 7–8B replication, T1, T2, T5, T6 and the half-LR test), whose files also hold our predictions. The 1B family-by-recipe test was written after 11 of its 12 values were known. We do not adjust for multiplicity. The exploratory analyses of §7 were run after the results and carry no test. The other comparisons are descriptive.

Table 13: When each claim's rule was fixed.

Claim	Status	Note
OLMo's partial erosion (Constant)	Descriptive	Seed 1101 chose the setting; the fresh seeds give $+45.9$ pp
The families differ at one recipe	Descriptive	The 20–55 pp partial window was fixed after the Qwen results
Scheduled erodes less than Constant (OLMo)	Prediction before the run	Held: $+28.8$ against $+45.1$ pp
Constant, token-mean gives $E \leq 36.25$ pp	Prediction before the run	Failed: $+41.9$ pp
Rank matters little (Scheduled, rank 16)	Rule before the run	One seed each
The recipe effect differs by family at 1B (OLMo against Llama)	Design before two seeds; exact test after 11 of 12 values	Excludes 0 on Qwen3Guard and on HarmBench
7–8B replication (OLMo-2-7B against Llama-3.1-8B)	Before nine new runs; three Constant values known	Not shown on both judges; point estimate reverses; two primary predictions failed, Llama prediction held
Scheduled on Qwen2.5-1.5B: T1, T2	Before new Qwen Scheduled runs; comparator cells known	T1 differs (OLMo) on both judges; T2 not shown on Qwen3Guard, differs on HarmBench; test predictions held, size prediction partly
Scheduled on Qwen2.5-7B: T5, T6	Before new Qwen Scheduled runs; comparators known or interim	T5 differs (Qwen) on Qwen3Guard only, clearing 0 narrowly; T6 not shown; T5 prediction failed
Half LR at 7–8B	Before new OLMo half-LR runs; Llama half-LR known; idea from interim data	Not shown on both judges; both predictions held; equality not established
The opening switch carries about half	Descriptive	Observational
Refusal direction, answer length, eroded prompts, baselines	Exploratory, after the results	No test; one to three seeds per cell
Prospective OLMo component pilot	Frozen before six new runs	One fresh seed; checkpoint-conditional descriptive recipe effects; no timing-versus-mean-rate or mediator identification
Blind human read	Sample, rubric and analysis before reading	One reader
Fiction: content and Dolly labels	Pre-registered, frozen	One timing deviation, recorded before any result
Family order at 7–8B; the three-seed Qwen rows	Descriptive	Three seeds each, on a later engine identity

G Prospective controlled component pilot

After the historical comparisons, we froze a separate one-seed diagnostic block on 12 October 2026, before its runs. It tests the two OLMo checkpoints at fresh seed 1121, with one engine (`472fc7f6`), the same 4,800 Dolly rows, 38 optimizer updates and the original development evaluation. Within each checkpoint, A/C/D start from the same recorded adapter initialization in fresh processes with reset optimizer state. The actual training-row order, safety sample layout, generation and judge settings match across all six units. Run order was randomized once before outcomes: D/A/C at 1B and C/D/A at 7B. Historical results are not pooled into this block.

A uses example-mean response loss and a constant learning rate 10^{-4}. C retains that loss and adds four-step warmup/cosine decay with peak 10^{-4} and observed 38-update mean 5×10^{-5}. D switches to token-mean loss under the same schedule. Thus C$-$A estimates the schedule package under example weighting, and D$-$C estimates token weighting under that schedule. Warmup, decay timing and the lower average learning rate remain bundled. The absent token-mean/constant-rate arm prevents a factorial loss-by-schedule interaction estimate.

The measured Unsafe-share contrasts C$-$A, D$-$C and D$-$A are -12.00, -8.50 and -20.50 pp at 1B, and $+7.50$, $+2.83$ and $+10.33$ pp at 7B. Both controlled component changes therefore contribute to the opposite recipe responses in this fixed block. The 1B-minus-7B contrasts are -19.50 and -11.33 pp; their sum is the total -30.83 pp difference. This is an ordered decomposition of interventions, not a representation-mediation estimate or a uniquely identified attribution. The additional 1B

Table 14: Prospective component pilot, seed 1121. Unsafe counts use all 600 sampled answers; no guard label was invalid. Benign refusals use 100 answers; ARC uses 189 forced-choice questions. Query NLL has the same example-mean reduction in every arm and is a learning indicator, not an independently verified held-out utility measure.

Model	Arm	Unsafe/600	Ref./100	ARC/189	NLL
OLMo-1B	A	301	3	83	1.542
OLMo-1B	C	229	3	85	1.570
OLMo-1B	D	178	19	74	1.617
OLMo-7B	A	242	1	169	1.354
OLMo-7B	C	287	1	169	1.384
OLMo-7B	D	304	5	169	1.417

reduction under token weighting accompanies 16 pp more benign refusal and 11 fewer correct ARC answers than C. It does not establish a selective-safety improvement.

The registration funds one independent training seed and prespecifies checkpoint-conditional descriptive contrasts. The 600 sampled answers are not training replicates; no population confidence interval, p-value or size law is reported. The checkpoints are different released models, so parameter count itself was not isolated. The experiment intervenes on recipe components, not on refusal directions or a pretraining mechanism. All six units and both successful batch receipts passed sealed-file, identity and pairing verification before comparison. Verification checks recorded identities and saved bytes; it does not prove remote numerical execution authenticity or independently audit all prepared source data. No secondary-judge result is claimed for this new block.

Protocol SHA-256 prefix: `084d5543`; reader: `bf7d49f9`; archived analysis: `b2643ca1`.

H Screening log

We list every setting we ran, in time order. All runs use LoRA rank 64, unless marked; §2 defines the recipes, and Section A gives every seed.

1. Qwen2.5-1.5B, High-dose, 3 seeds: mean $E = +81.8$ pp after 3 epochs. It failed the bar that answers must end (repetition loops).

2. Qwen2.5-1.5B, Constant, seed 1101: $+75.2$ pp. Seed 1102 was lost to a timer and gave no result.

3. Qwen2.5-7B, Constant, seed 1101: $+82.2$ pp.

4. Qwen2.5-1.5B, Constant, rank 16, seed 1101: $+67.8$ pp.

5. OLMo-2-1B, Constant, seed 1101: $+43.5$ pp; then seeds 1102 and 1103.

6. OLMo-2-1B, Scheduled, seed 1101: $+29.0$ pp. We pre-registered the prediction that it erodes less than the Constant recipe.

7. Llama-3.2-1B, Constant, seed 1101: $+62.5$ pp.

8. OLMo-2-1B, Scheduled, seeds 1102 and 1103; Llama-3.2-1B, Scheduled, seeds 1101–1103; OLMo-2-1B, Constant, token-mean and Scheduled, rank 16; both families under Scheduled, half LR (Table 2).

9. A Qwen2.5-1.5B Scheduled run was planned for this note, with a written prediction, and dropped before any run (9 Oct). Nothing from it exists.

10. Llama-3.2-1B, Constant, seeds 1102 and 1103 (9 Oct): $+61.3$ and $+54.0$ pp. Seed 1102 needed a second attempt: a cap gate stopped the first attempt before its job started.

11. Llama-3.2-1B, Constant, half LR, seed 1101 (9 Oct): $+48.0$ pp. Granite, judged on the pod, agrees in direction (greedy E $+47.5$ pp).

12. OLMo-2-1B, Constant, token-mean, seeds 1102 and 1103 (9–10 Oct), to complete three seeds: $+39.5$ and $+42.5$ pp.

13. OLMo-2-1B, Constant, on dark and on light fiction instead of Dolly, seeds 1101–1103 each (9–10 Oct), pre-registered (§8): every unit is reported.

14. On a later engine identity (10 Oct): OLMo-2-7B and Llama-3.1-8B, Constant, seed 1101 ($+49.7$ and $+70.2$ pp), then OLMo-2-7B seed

1102 (+37.2); Qwen2.5-1.5B and Qwen2.5-7B, Constant, seeds 1101–1103, where seed 1101 reproduced the earlier units' counts exactly; Llama-3.1-8B, Constant, half LR, seeds 1101 and 1102 (+52.5 and +54.0 pp), run as the larger study's second learning-rate rung.

15. Pre-registered (10–11 Oct; §5): the 7–8B replication (OLMo-2-7B Constant seed 1103, Llama-3.1-8B Constant seeds 1102 and 1103, and Scheduled seeds 1101–1103 for both); Qwen2.5-1.5B and Qwen2.5-7B, Scheduled, seeds 1101–1103; OLMo-2-7B, Constant, half LR, seeds 1101–1103. Every unit is reported.

16. Refusal-direction probes (exploratory; §7) on the 1B and 7–8B Constant and Scheduled units; 20 of 24 planned units were recovered by 12 Oct, and all 20 are reported. The earlier 13-unit snapshot is preserved; four units remain unavailable.

17. A separate prospective component pilot (12 Oct; Section G): OLMo-1B and OLMo-7B, three recipes at fresh seed 1121, on one engine. All six units are reported separately from the historical seed means.

18. A subsequent follow-up at seed 1122 attempted a matched-mean learning-rate comparison and a direct activation intervention. Startup, resource and retrieval failures left no locally verified paired comparison or activation result. These attempts do not enter the reported contrasts; their absence does not support a null effect.

Judge wording, applied. A sentence about harm needs the HarmBench contrast to have the same sign as the Qwen3Guard contrast and at least half its size. If E_{HB} is below 20 pp, no sentence says harm. For OLMo-2-1B under the Constant recipe the rule holds on every seed ($E_{\mathrm{HB}} \geq 20$ pp and at least half of E), so we say harm there, meaning harmful compliance as HarmBench defines it: a blind human read found the HarmBench bar looser than "usable help" (4 of 12 HarmBench *yes* answers were rated usable; §9). Under the Scheduled recipe, E_{HB} is below 20 pp on every seed, so we use lost-refusal wording, although the same-sign, half-size condition holds. This wording threshold does not establish absence of harmful compliance. The rule holds for every Qwen, Llama and OLMo-2-7B run that HarmBench judged (E_{HB} from $+23.0$ to $+63.5$ pp, at least 0.59 of E), so we say harm there too. In both fiction arms E_{HB} is below 20 pp, so we say lost refusal there.

Qwen2.5-1.5B recipe screens. Earlier screens suggest that Qwen2.5-1.5B responds little to other recipe settings: cutting the dose about 30 times, from the High-dose recipe after one epoch (600 steps at 2×10^{-4}) to the Constant recipe, moved U from about 83% to 80%, and Constant, rank 16 gives $+67.8$ pp, about 7 pp below rank 64 (Section A). The Scheduled recipe does not lower its erosion either (§5).